

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**3rd Semester of B. Tech (EE) Examination****March 2018****EE 241 ANALOG & DIGITAL ELECTRONICS****Date: 30/03/2018, Friday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1 Answer the questions below.****[07]**

- a. The number of depletion layer in transistor is_____.
(i) four (ii) three
(iii) one (iv) two
- b. The base of transistor is_____doped.
(i) heavily (ii) moderately
(iii) lightly (iv) none of the above
- c. The pinch-off voltage of a JFET is about_____.
(i) 5V (ii) 0.6V
(iii) 15V (iv) 25V
- d. A MOSFET is sometimes called_____JFET.
(i) many gate (ii) open gate
(iii) insulated gate (iv) shorted gate
- e. Define input offset current.
- f. Define common mode rejection ratio.
- g. Define slew rate.

Q – 2.a What is an Op-Amp? Give the advantages of Op-Amp over conventional amplifier. **[04]**
Draw also its pin diagram.

Q – 2.b Answer any two questions. **[10]**

- (i) Explain OP-AMP is used as averaging amplifier using with necessary diagram.
- (ii) Explain Differentiator circuit using Op-amp.
- (iii) Give the difference between JFET and BJT. Also write the applications of JFET.

Q – 3 Answer any two questions. [14]

- Give the types of comparator. Write applications of comparator. Explain the working of non inverting comparator with circuit diagram and output wave forms.
- Define active and passive filters. Explain the first order low pass filter with circuit diagram.
- Define stabilization and stability factor. Also derive equation for stability factor.

SECTION – II

Q – 4 Answer the questions below. [07]

- Write the examples of digital system.
- Write the applications of flip-flop.
- Do the binary addition of $A = (10111)_2$ & $B = (11001)_2$.
- Find 1's complement of $(10010100)_2$.
- Do the binary subtraction of $A = (11011)_2$ from $B = (10110)_2$.
- Find the 2's complement of $(10111000)_2$.
- Write the advantages of digital system.

Q – 5.a Explain the multiplexer. Also give advantages of multiplexer. [05]

OR

Q – 5.a Explain the working of a half adder. Draw its logic diagram. [05]

Q – 5.b Give the classification & numbers of components is used for logic families. [04]

Q – 5.c Explain the working of diode transistor logic with the help of neat circuit diagram. [05]

OR

Q – 5.c Explain the working of CMOS inverter logic with the help of neat circuit diagram. [05]

Q – 6.a Reduce the following expressions using K-Map & draw logic diagram. [07]

(i) $\sum m(1,2,9,10,11,14,15)$

(ii) $\sum m(0,1,2,5,13,15)$

Q – 6.b (i) Convert $(85.63)_{10}$ number into equivalent binary number. [05]

(ii) Convert the binary $(11010010)_2$ number into equivalent octal number.

(iii) Convert $(436)_8$ number into equivalent hex number.

Q – 6.c Perform $(4)_{10} - (9)_{10}$ using 1's complement method. [02]

OR

Q – 6.a Explain the concept of level triggering and edge triggering. **[07]**

Q – 6.b Draw a logic circuit from Boolean equation. **(Any Two)** **[04]**

(i) $Y = AB + BC + \bar{A}\bar{B}$

(ii) $Y = (A + B)(A + C)$

(iii) $Y = \bar{A}B\bar{C} + D$

Q – 6.c Draw different logic gates with their Boolean expression. **[03]**
